Database Management System (DBMS)

Previous Year Question Paper – 2021
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. Explain the term DBMS and list its advantages over a file processing system. *(5 marks each)*

Q2. Describe data independence and explain why it is important in DBMS. *(5 marks each)*

Q3. What is the purpose of normalization? Illustrate 1NF, 2NF and 3NF using an example table. *(5 marks each)*

Q4. Explain Atomicity, Consistency, Isolation and Durability in the context of transactions. *(5 marks each)*

Q5. Distinguish between a primary key, a candidate key and a super key. *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. Explain functional dependency and how it is used to determine normal forms. *(10 marks each)*

Q2. Classify SQL commands into DDL, DML and DCL with suitable examples. *(10 marks each)*

Q3. Describe different types of joins used to combine data from multiple tables. *(10 marks each)*

Q4. Why is indexing important in DBMS? Compare B-Tree and B+ Tree indexes. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Model a University Management System with Students, Courses and Faculty, and write sample SQL queries for common operations. *(15 marks each)*

Q2. What is data abstraction in DBMS? Explain its levels with examples. *(15 marks each)*

Q3. Discuss how a DBMS recovers from a system failure using transaction logs. *(15 marks each)*

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